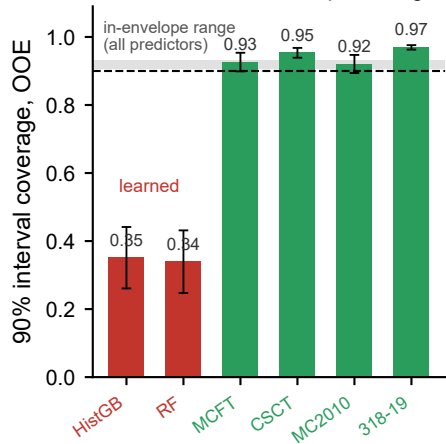
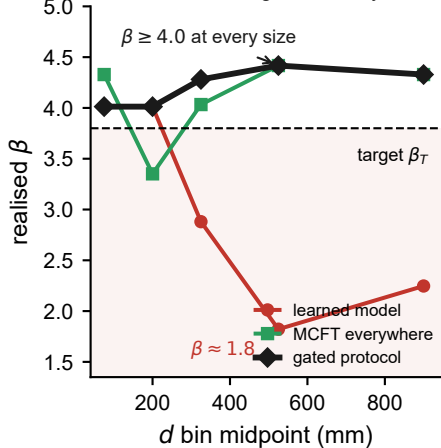


A Same pipeline, same metric:
mechanical models keep coverage



B Envelope gate + fallback
restores target reliability



C Physics-anchored learning helps;
the gate is still needed

